Customer Review Sentiment Analysis (NLP)

An end-to-end NLP project that classifies customer reviews of a smartphone as **Positive**, **Neutral** or **Negative**, and serves the best model through an interactive **Streamlit** web app.

Live demo: [add your Streamlit app link here]

Business objective

Extract sentiment from customer reviews of a product, so that a business can quickly see how customers feel without reading every review.

Dataset

- 1,440 customer reviews in `dataset.xlsx`
- Columns: `title`, `rating` (1-5 stars) and `body` (review text)
- No missing values and no duplicate rows
- Sentiment labels are created from the star rating:
  - 1-2 stars: **Negative**
  - 3 stars: **Neutral**
  - 4-5 stars: **Positive**
- After cleaning, 1,436 reviews remain: 729 Positive, 508 Negative and 199 Neutral

Project workflow

1. **Exploratory data analysis (EDA):** rating distribution, review length in characters and words, most frequent words, and word clouds for positive and negative reviews.
2. **Text cleaning:**
   - detect the language of each review
   - translate non-English reviews to English
   - convert to lowercase and remove links, HTML tags, non-ASCII characters, punctuation and numbers
   - remove stop words and short words, then lemmatize with NLTK
3. **Feature extraction:** TF-IDF with the top 5,000 features.

4. **Model building:** an 80/20 stratified train-test split (1,148 training and 288 test reviews), then three classifiers trained and compared.
5. **Evaluation:** accuracy, precision, recall, F1 score and confusion matrices.
6. **Deployment:** the best model and the TF-IDF vectorizer are saved with `joblib` and served through a Streamlit app.

Results

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.743	0.711	0.743	0.692
Random Forest	0.733	0.776	0.733	0.680
Linear SVM (best)	**0.757**	0.734	**0.757**	**0.733**

Precision, recall and F1 are weighted averages across the three classes. Linear SVM had the highest F1 score, so it was selected for the app.

Project structure

```
.
├── NLP_PROJECT.ipynb        # EDA, cleaning, model building and evaluation
├── dataset.xlsx             # customer reviews
├── app.py                   # Streamlit web app
├── requirements.txt         # packages needed to run the app
├── sentiment_model.pkl      # saved Linear SVM model
├── tfidf_vectorizer.pkl     # saved TF-IDF vectorizer
└── README.md
```

How to run

1. Clone the repository

```
git clone https://github.com/venugopi3232/<your-repo-name>.git
cd <your-repo-name>
```

2. Install the packages for the app

```
pip install -r requirements.txt
```

3. Start the app

```
streamlit run app.py
```

Then type a customer review in the text box and click **Predict Sentiment**.

To re-run the notebook, also install:

```
pip install pandas numpy matplotlib seaborn wordcloud langdetect deep-
translator openpyxl
```

Tech stack

Python, Pandas, NumPy, NLTK, Scikit-learn, TF-IDF, Matplotlib, Seaborn, WordCloud, Streamlit, Joblib, Git and GitHub.

Limitations and future improvements

- The model rarely predicts the **Neutral** class (F1 of about 0.26 for Linear SVM), because Neutral reviews are fewer and use words similar to both Positive and Negative reviews.
- Planned improvements:
  - balance the classes or use class weights for Linear SVM
  - try n-grams and hyperparameter tuning
  - try transformer-based models such as BERT
  - collect more reviews to improve Neutral predictions

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